

AI Cart Showcase

Introduction to Artificial Intelligence IFSA



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Presented by

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Agent Overview

```
class hiyori_brain:
    """One-hidden-layer MLP racing agent for AI Cart V2.

    The network receives the nine engine raycasts plus features derived only
    from those raycasts and from ``pascardata``. Its four sigmoid outputs
    map directly to throttle, brake, left, and right.
    """

    def __init__(self) -> None:
        self.score = BASE_FITNESS
        self.decider = 0
        self.x = 0.0
        self.y = 0.0
        self.speed = 0.0
        self.chars = string.ascii_letters + string.digits

        self.init_param()

    def init_param(self) -> None:
        # Xavier-style initialization keeps sigmoid activations in a useful
        # range while still providing diverse agents for evolutionary search.
        limit1 = np.sqrt(6.0 / (N_INPUTS + N_HIDDEN))
        limit2 = np.sqrt(6.0 / (N_HIDDEN + N_ACTIONS))
        self.W1 = np.random.uniform(-limit1, limit1, (N_HIDDEN, N_INPUTS))
        self.b1 = np.zeros(N_HIDDEN, dtype=float)
        self.W2 = np.random.uniform(-limit2, limit2, (N_ACTIONS, N_HIDDEN))
```



I used a MLP (Multilayer Perceptron) neural network.

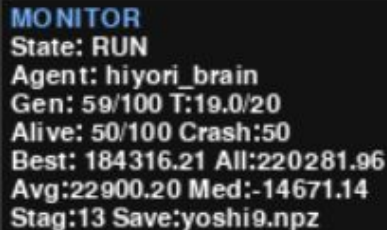
- 1 hidden layer with 16 nodes
- Trained on multiple maps
- Generative AI tools used

calculate_score()

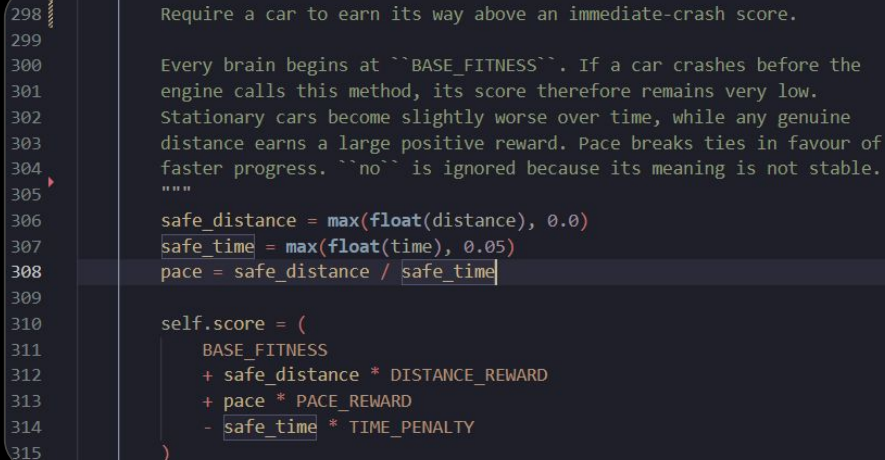
Negative starting score — heavily penalizes cars that stay at the starting line.

Score decrease for no movement.
Increasing total distance traveled increases score.

Speed used as a tiebreaker for cars that travel the same distance.



```
MONITOR
State: RUN
Agent: hiyori_brain
Gen: 59/100 T:19.0/20
Alive: 50/100 Crash:50
Best: 184316.21 All:220281.96
Avg:22900.20 Med:-14671.14
Stag:13 Save:yoshi9.npz
```



```
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    """
    Require a car to earn its way above an immediate-crash score.

    Every brain begins at ``BASE_FITNESS``. If a car crashes before the
    engine calls this method, its score therefore remains very low.
    Stationary cars become slightly worse over time, while any genuine
    distance earns a large positive reward. Pace breaks ties in favour of
    faster progress. ``no`` is ignored because its meaning is not stable.
    """

    safe_distance = max(float(distance), 0.0)
    safe_time = max(float(time), 0.05)
    pace = safe_distance / safe_time

    self.score = (
        BASE_FITNESS
        + safe_distance * DISTANCE_REWARD
        + pace * PACE_REWARD
        - safe_time * TIME_PENALTY
    )
```

Difficulties

- Keeping track of distance
 - Engine glitches
 - Distance traveled $\neq$ race completion
 - No reward for race completion
- Mutation rate vs. rate of kept cars
- Throttle activation



I went through multiple iterations of code.

Thank you

Any questions or
comments?

Repo link:

[https://github.com/yokunerukosbelt/AI_CA
RT_engine_PUBLIC_EN](https://github.com/yokunerukosbelt/AI_CA_RT_engine_PUBLIC_EN)